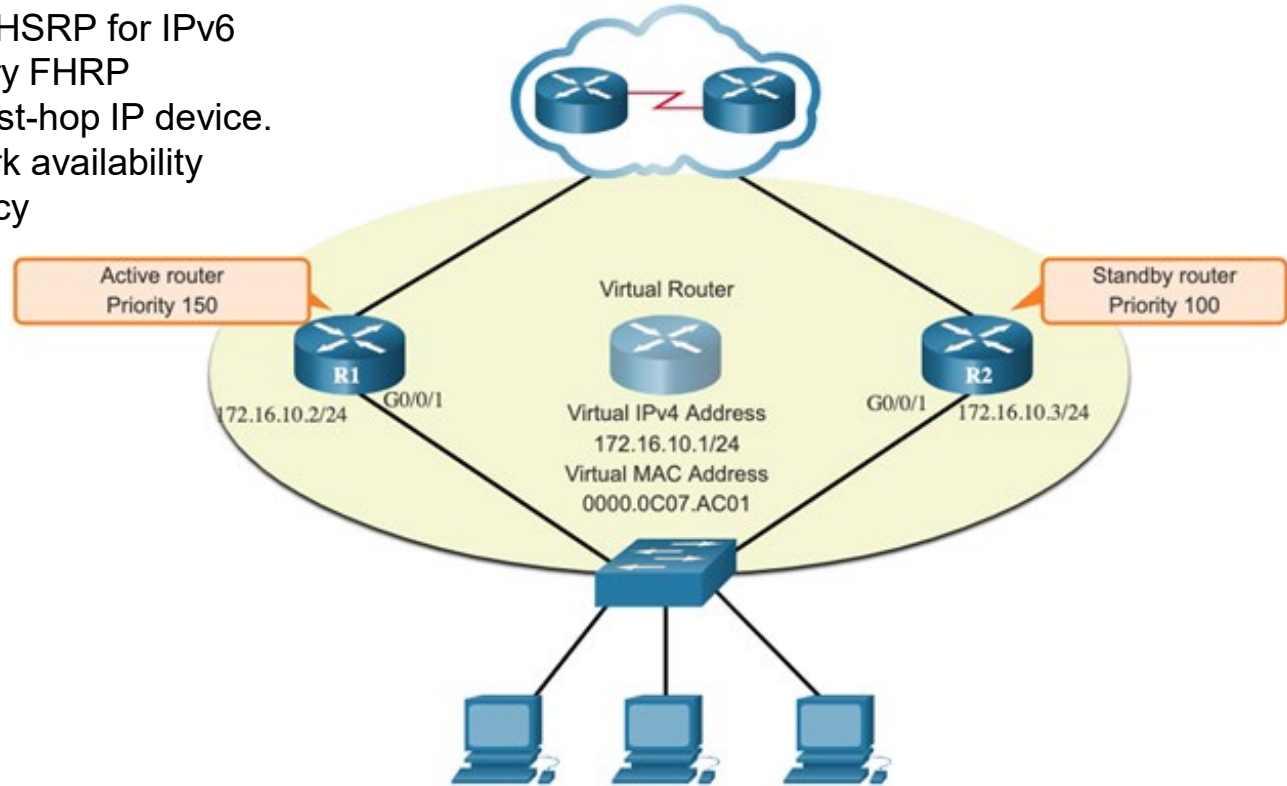


# 9.2 HSRP

# HSRP Overview

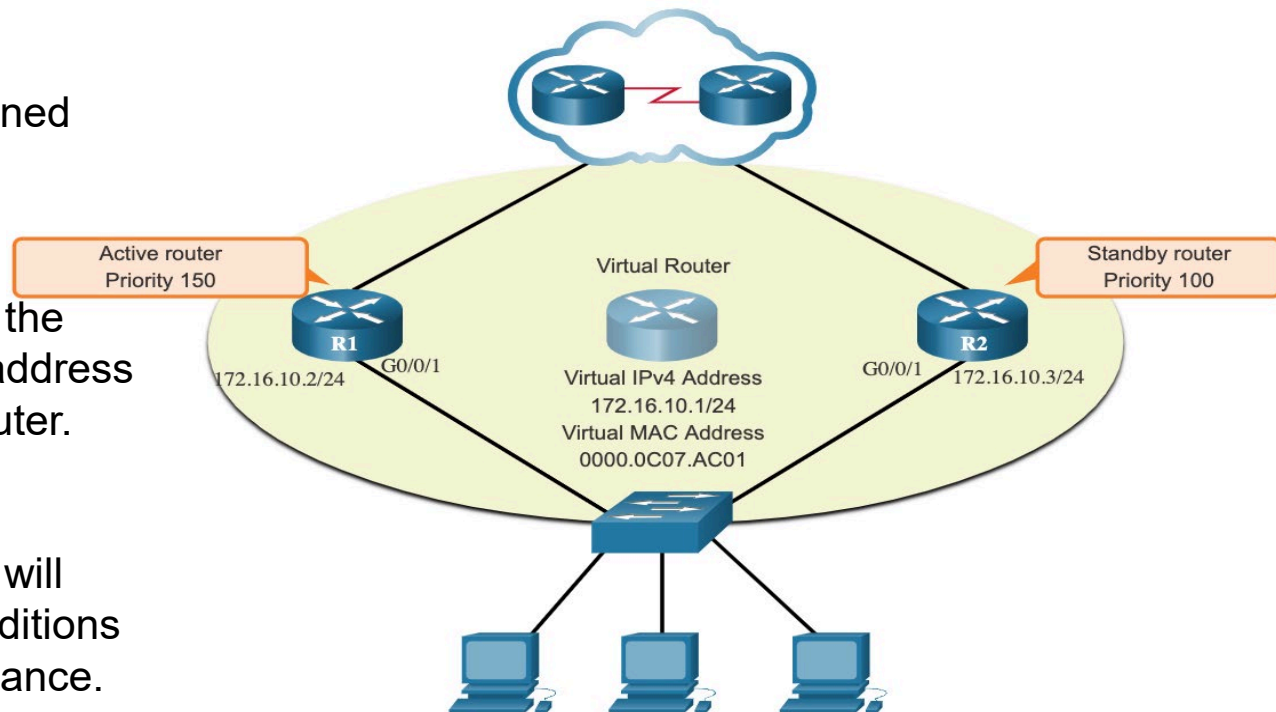
- Cisco provides HSRP and HSRP for IPv6
- HSRP is a Cisco-proprietary FHRP
- Transparent failover of a first-hop IP device.
- HSRP ensures high network availability
- First-hop routing redundancy



# HSRP

## HSRP Priority and Preemption

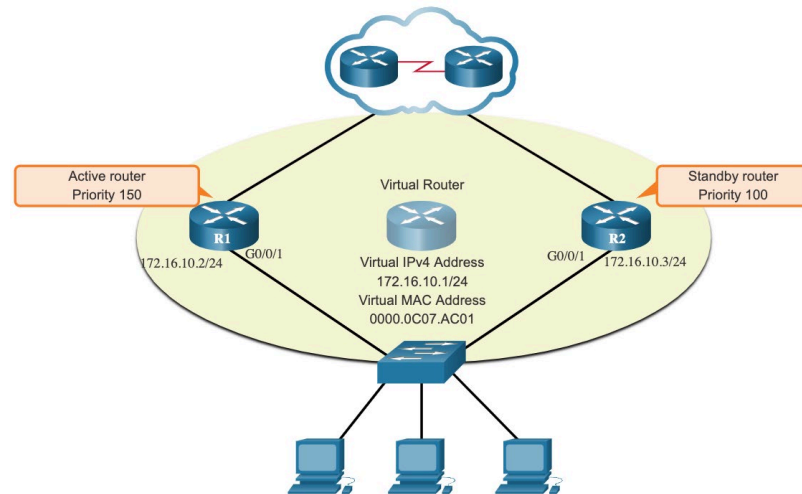
- The role of the active and standby routers is determined during the HSRP election process.
- By default, the router with the numerically highest IPv4 address is elected as the active router.
- Control how your network will operate under normal conditions rather than leaving it to chance.



# HSRP

## HSRP Priority and Preemption

- HSRP priority can be used to determine the active router.
- The router with the highest HSRP priority will become the active router.
- By default, the HSRP priority is 100.
- If the priorities are equal, the router with the numerically highest IPv4 address is elected as the active router.
- To configure a router to be the active router, use the **standby priority** interface command. The range of the HSRP priority is 0 to 255.

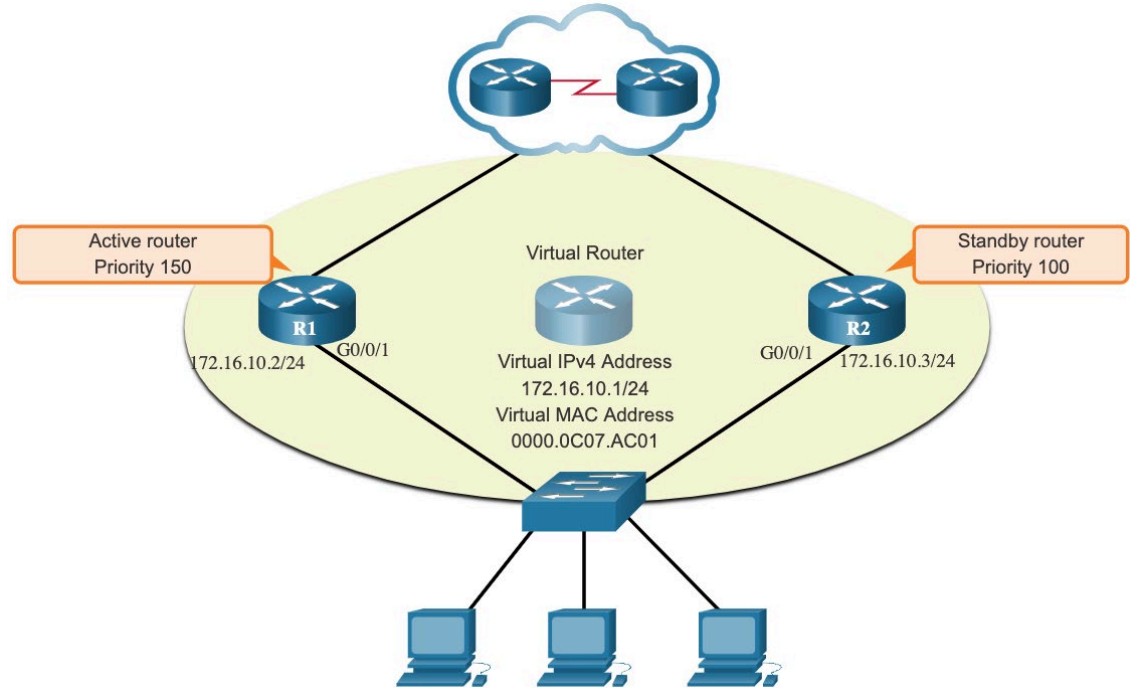


# HSRP

## HSRP Priority and Preemption (Cont.)

By default, after a router becomes the active router, it will remain the active router even if another router comes online with a higher HSRP priority.

Preemption is the ability of an HSRP router to trigger the re-election process.



# HSRP

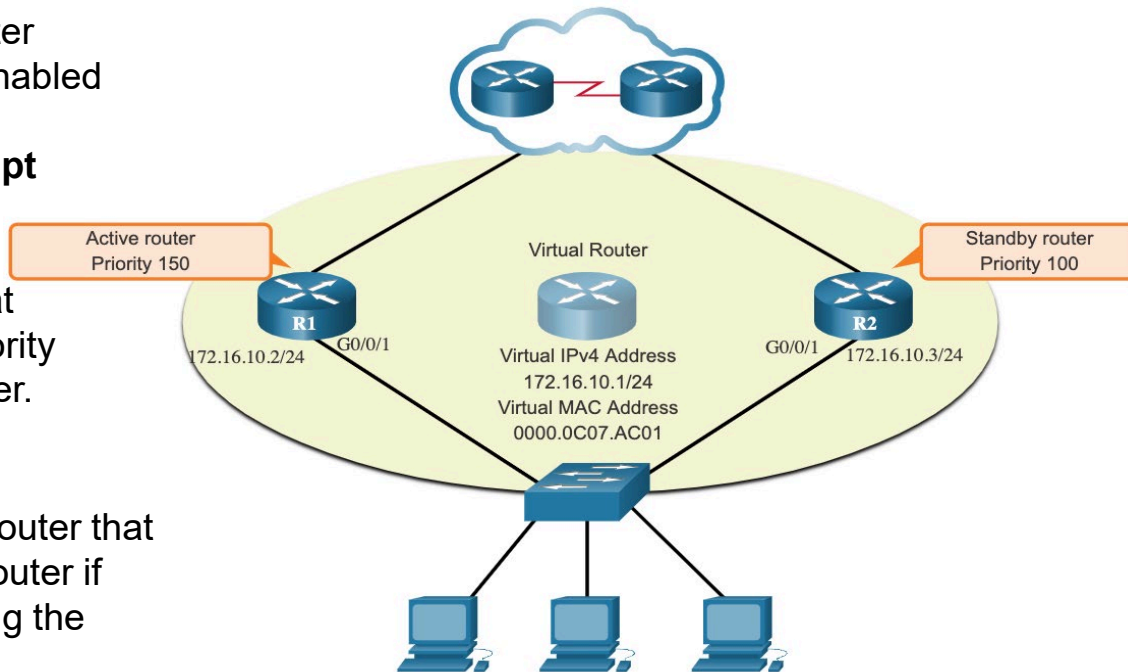
## HSRP Priority and Preemption (Cont.)

- To force a new HSRP election process to take place when a higher priority router comes online, preemption must be enabled using the interface command.

**Router(config-if)# standby preempt**

- With preemption enabled, a router that comes online with a higher HSRP priority will assume the role of the active router.

**Note:** With preemption disabled, the router that boots up first will become the active router if there are no other routers online during the election process.



# HSRP

## HSRP States and Times

- The active and standby HSRP routers send hello packets to the HSRP group multicast address every 3 seconds by default.
- The standby router will become active if it does not receive a hello message from the active router after 10 seconds. (Hold Timer)
- You can lower these timer settings to speed up the failover or preemption.
- To avoid increased CPU usage and unnecessary standby state changes,
  - Minimum Hello timer 1 second
  - Minimum Hold timer 4 seconds.

# HSRP

## HSRP States and Times

| HSRP State | Description                                                                                                                                                                       |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Initial    | This state is entered through a configuration change or when an interface first becomes available.                                                                                |
| Learn      | The router has not determined the virtual IP address and has not yet seen a hello message from the active router. In this state, the router waits to hear from the active router. |
| Listen     | The router knows the virtual IP address, but the router is neither the active router nor the standby router. It listens for hello messages from those routers.                    |
| Speak      | The router sends periodic hello messages and actively participates in the election of the active and/or standby router.                                                           |
| Standby    | The router is a candidate to become the next active router and sends periodic hello messages.                                                                                     |